AI FOR CREATIVITY · LEVEL 6 · FUTURE EDUCATION UNIVERSITY COLLEGE

Threads of Memory

Using transfer learning to tell hand-woven Al Sadu from machine-printed imitations — an annotated exemplar R&D document

Item	Detail
Student	Student A (BA Fashion & Textiles, exemplar persona)
Unit	Artificial Intelligence for Creativity · Level 6 · Portfolio component 1: R&D document
Learning objectives evidenced	LO1 (contexts, concepts, processes) · LO3 (ethical, refined application)
Word count (excluding tables, references, appendices)	About 3,000

How to read this exemplar

This is a **fictional, illustrative** R&D document written by your lecturer to show what a strong submission looks like. The student, the search counts, the dataset numbers and the results are **invented for teaching**. The published sources in the reference list are real.

Green boxes explain **why a section works**. Red boxes flag **common mistakes** to avoid. Your own document will be about your own project — use the structure, not the content.

Abstract

Al Sadu, the Bedouin weaving tradition of the United Arab Emirates, was inscribed on UNESCO's List of Intangible Cultural Heritage in Need of Urgent Safeguarding in 2011 and moved to the Representative List in 2025 (UNESCO, 2025). As Sadu-style patterns become popular in fashion and interiors, cheap machine-printed imitations are sold alongside hand-woven pieces, and buyers often cannot tell them apart. This project investigates whether a small, ethically collected image dataset and transfer learning can distinguish hand-woven Al Sadu from machine-printed Sadu-style fabric and from other woven textures. I built a 1,230-image dataset, fine-tuned a pretrained ResNet-18 using fastai, and reached 86% accuracy on a test set of objects the model had never seen. The project's contribution is less the accuracy than the process: a documented, consent-based dataset of Emirati craft, and an honest account of how data leakage first made my results look better than they were.

✔️ TUTOR NOTE — the abstract

It answers five questions in one paragraph: **context** (why Sadu matters), **problem** (imitations), **what I did** (dataset + method), **result** (86% on unseen objects) and **contribution** (process and dataset, stated modestly).

Every factual claim about the world is cited — here the UNESCO dates.

1. Introduction and context

Al Sadu is a form of weaving traditionally practised by Bedouin women, using sheep, goat and camel wool on a ground loom to create geometric patterns in strong reds, blacks and whites (UNESCO, 2025). UNESCO's inscription in 2011 recognised both its cultural importance and the risk that the skills were declining; its transfer to the Representative List in 2025 reflects safeguarding work since then (UNESCO, 2025).

In my own practice as a textile designer I have seen Sadu motifs printed onto polyester for cushions, bags and abayas. These prints can be beautiful, but when they are sold as 'Sadu' they blur the line between a craft and its imitation, and they return nothing to the weavers. This project began with a practical question from a weaver I interviewed: *can a phone tell a buyer whether a piece is really hand-woven?*

The question sits at the meeting point of three contexts: **cultural heritage** (safeguarding a living craft), **creative industries** (fashion and interiors borrowing heritage motifs), and **AI** (computer vision models that can learn visual textures from examples).

✔️ TUTOR NOTE — the introduction

It moves from the **world** (heritage) to the **practice** (the student's own discipline) to the **question**. That is what 'contextualise your practice' means.

A personal starting point (the weaver's question) is welcome in creative R&D — as long as it is followed by evidence.

⚠️ WATCH OUT — unsupported claims

Avoid sentences like 'AI is revolutionising every industry' or 'everyone knows fakes are everywhere'. If you cannot cite it or show it, cut it or soften it ('In my own observation…').

2. Review of literature and practice

2.1 Creativity and computation

Boden (2004) distinguishes combinational, exploratory and transformational creativity. My project is not an attempt to make a machine creative; it uses AI as a tool to protect human creativity. Boden's framework is still useful, because it helps me argue that the creative value of Al Sadu lies in the

weaver's exploration of a traditional pattern space — something a printed copy reproduces visually but does not perform.

2.2 Recognising textures with deep learning

Cimpoi *et al.* (2014) showed that textures can be described with human-understandable attributes such as 'woven', 'braided' and 'zigzagged', and released the Describable Textures Dataset (DTD) of 5,640 images in 47 categories. Their work suggests that texture — not just shape — carries information a model can learn, which matters for fabric.

Training a deep network from scratch needs very large datasets such as ImageNet (Deng *et al.*, 2009). Transfer learning offers an alternative: Yosinski *et al.* (2014) found that features learned in early layers of a network are general and transfer well to new tasks. Howard and Gugger (2020) show how a pretrained residual network (He *et al.*, 2016) can be fine-tuned in fastai with a few hundred images per class. This makes a small, carefully collected craft dataset realistic for a student project.

2.3 Data, bias and documentation

Buolamwini and Gebru (2018) demonstrated that commercial vision systems performed much worse on under-represented groups, showing how gaps in training data become gaps in performance. Gebru *et al.* (2021) propose 'datasheets for datasets' so that creators record how data was collected, by whom and for what purpose; Mitchell *et al.* (2019) propose similar 'model cards' for trained models. I use both ideas in Section 5 and Appendix D.

2.4 Synthesis: literature matrix

Source	What it contributes	Limitation for my project
Boden (2004)	Framework for types of creativity	Not about AI tools that support craft
Cimpoi *et al.* (2014)	Textures are learnable; public texture dataset	No Arabian Gulf textiles; no hand-made vs printed distinction
Yosinski *et al.* (2014); He *et al.* (2016)	Pretrained features transfer to new tasks	Tested on general objects, not heritage craft
Howard and Gugger (2020)	Practical fine-tuning with small data	Tutorial examples, not evaluation on cultural data
Buolamwini and Gebru (2018)	Under-representation causes performance gaps	About faces, but the principle transfers to culture
Gebru *et al.* (2021); Mitchell *et al.* (2019)	How to document datasets and models	Frameworks, not applied to craft

✔️ TUTOR NOTE — the review

It is organised by **theme**, not by source ('Smith said… Jones said…'). Each paragraph ends by saying what the sources mean **for this project**.

The literature matrix turns reading into analysis: the right-hand column is where the research gap starts to appear.

⚠️ WATCH OUT — summarising without synthesising

A list of summaries is not a review. For every source ask: *so what, for my project?* and *what does it NOT cover?*

3. Identifying the research gap

To check whether my question had already been answered, I ran a structured search on 12/09/2026 using Google Scholar, the UCA library catalogue and the CVF Open Access archive. The counts below are recorded in my search log (Appendix B).

Search string	Results screened	Relevant	What I found
"textile classification" AND "deep learning"	First 50	14	Mostly industrial defect detection and garment types
"Sadu" OR "Al Sadu" AND ("machine learning" OR "computer vision")	All 9	0	Cultural and historical studies only
"handwoven" AND ("counterfeit" OR "imitation") AND "image"	All 23	2	Handloom authentication in South Asia; no Gulf textiles
"cultural heritage" AND "textile" AND "dataset"	First 50	5	Museum digitisation; no labelled datasets for Sadu

From the matrix and search log I identify three gaps:

1. **Context gap** — texture and textile recognition research rarely includes Arabian Gulf or Emirati textiles.
2. **Data gap** — I found no public, labelled, consent-based image dataset of Al Sadu.
3. **Application gap** — distinguishing hand-made craft from printed imitations has been explored for other regional crafts, but not for Al Sadu.

✔️ TUTOR NOTE — the gap

The gap is **evidenced**, not asserted: search strings, dates, databases, counts and a log in the appendix. Another student could repeat the search.

The gap is named by **type** (context, data, application). This makes it precise and easy for an assessor to recognise.

⚠️ WATCH OUT — fake gaps

'Nobody has ever done this' is almost never true and cannot be proved. Say 'In my search of X, Y and Z on [date], I found no…' — that is honest and checkable.

A gap is not a gap if nobody needs it filled. Link it to a real problem (Section 4).

4. Problem statement, aim and research questions

Problem statement

Machine-printed Sadu-style fabrics are sold alongside hand-woven Al Sadu, and buyers cannot easily tell them apart. This undermines weavers' income and the value of a safeguarded craft. Existing texture-recognition research and datasets do not include Emirati textiles, so there is no tested tool or dataset to support authentication.

Aim

To explore whether transfer learning on a small, ethically collected image dataset can distinguish hand-woven Al Sadu from machine-printed imitations and other woven textures, and to document the dataset and process so others can build on it.

Research questions

1. **RQ1** How accurately can a fine-tuned pretrained CNN classify hand-woven Al Sadu, machine-printed Sadu-style fabric and other woven textures, on objects it has not seen?
2. **RQ2** Which visual features does the model rely on, and where does it fail?
3. **RQ3** What ethical and practical conditions must be met to collect and share a craft dataset responsibly?

Objectives

- Collect and document at least 350 images per class with written consent by Week 8.
- Train and evaluate a baseline model by Week 10; improve it and analyse errors by Week 12.
- Produce a datasheet and model card, and a Gradio prototype for user testing in Week 13.

✔️ TUTOR NOTE — the problem → aim → questions chain

Problem statement formula: **who** is affected + **what** is wrong + **why existing work doesn't solve it**.

Each research question is answerable with the planned method, and RQ3 makes ethics part of the research, not an afterthought.

⚠️ WATCH OUT — weak research questions

Too broad: 'Can AI help culture?' · Yes/no only: 'Can AI recognise Sadu?' · Not answerable in 14 weeks: 'Will this stop counterfeiting in the UAE?'

5. Dataset description (datasheet)

Following Gebru *et al.* (2021), this section documents why, how and under what conditions the data was collected.

Field	Description
Name and version	Threads-of-Memory v1.0 (September–October 2026)
Purpose	To train and evaluate a classifier that distinguishes hand-woven Al Sadu from machine-printed Sadu-style fabric and other woven textures, for research and teaching only.
Classes and counts	handwoven_sadu: 400 images of 58 textile pieces · printed_sadu_style: 350 images of 47 pieces · other_woven: 480 images (DTD) · Total 1,230
Sources	(1) My own photographs of hand-woven pieces at a community weaving workshop in Sharjah, with written permission from the organisers and each weaver whose work is shown. (2) My own photographs of printed fabrics bought from retail shops (receipts kept). (3) 120 images each from the DTD categories woven, braided, knitted and zigzagged (Cimpoi *et al.*, 2014), used under the dataset's terms for non-commercial research, checked on 10/09/2026.
Collection method	Phone camera (12 MP), daylight, fabric flat, 30 cm distance, 5–8 photos per piece from different areas and angles. No people or faces are included.
Labelling	Labelled by me at collection time; 60 random images re-checked by a weaver (agreement 58/60; 2 relabelled).

Field	Description
Pre-processing	Resized to 224 × 224 px; random crops, flips and small rotations as augmentation during training only.
Splits	Split **by piece, not by photo**: 70% train / 15% validation / 15% test, so all photos of one textile stay in the same split.
Licence and sharing	My photographs: CC BY-NC 4.0, shared only with consent-form permission. DTD images are not redistributed — users are directed to the original source.
Known limitations and bias	Only one workshop, so weaving styles from other emirates are under-represented; printed fabrics from 3 shops only; lighting is mostly daylight; the 'other woven' class comes from a different camera and source, which the model may learn as a shortcut.
Storage and privacy	Stored on university OneDrive; weavers' names kept separately from images; no personal data in file names.

✔️ TUTOR NOTE — the datasheet

It answers what an assessor needs: **what** the data is, **where** it came from, **who agreed**, **how** it was labelled and split, **what licence** applies and **what might be wrong with it**.

Numbers are specific (58 pieces, 400 images). 'A few hundred images from the internet' would score poorly.

Naming known biases is a strength, not a weakness — it shows critical thinking.

⚠️ WATCH OUT — dataset red flags

Images scraped from Instagram, Pinterest or Google Images with no permission or licence.

Splitting by photo instead of by object — near-identical photos end up in train and test (data leakage).

No mention of licence, consent or limitations.

6. Method and development log

I used fastai with a ResNet-18 pretrained on ImageNet (He *et al.*, 2016; Howard and Gugger, 2020), trained in Google Colab on a GPU, and tracked each experiment in a table so that changes and results can be compared.

Run	Date	Change	Validation accuracy	What I learned
1	02/10	Baseline: ResNet-18, 4 epochs, photos split randomly	94%	Suspiciously high — checked the split
2	05/10	Re-split by piece, not photo	78%	Run 1 had leakage: the model memorised individual textiles
3	08/10	Added augmentation; 8 epochs	82%	Better generalisation; printed vs hand-woven still confused
4	12/10	Added 110 close-up photos of printed fabric	86%	Close-ups show the flat, uniform print dots the model needed

On the held-out test set (185 images from 26 pieces never seen in training), the final model reached 86% accuracy. Most errors (19 of 26) were high-resolution printed fabrics predicted as hand-woven.

✔️ TUTOR NOTE — the development log

Failure is documented, not hidden. Run 1's 94% was wrong, and explaining **why** (leakage) is excellent evidence of learning and refinement (LO3).

Every change is one variable at a time, with a reason and a result.

⚠️ WATCH OUT — results without evidence

Always say which data the number comes from (validation or test) and include the confusion matrix or example errors in the appendix.

Do not screenshot a single lucky run — keep the log.

7. Ethical considerations

- **Consent and credit:** each weaver signed a consent form (Appendix C) covering photography, training use and optional credit by name. Two weavers chose not to be named.
- **Benefit and harm:** the tool could help buyers, but it could also be misused to label genuine work as fake. The prototype therefore shows a confidence score and a message that it is a research tool, not a certificate.
- **Cultural respect:** I discussed the project's framing with the workshop organisers and removed one motif they asked me not to publish.
- **Data protection:** no faces or personal data; storage follows university guidance.

✔️ TUTOR NOTE — the ethics section

Ethics is specific to **this** project and shows actions taken, not just principles. That is what the descriptor means by an 'ethical application' (LO3).

8. Reflection and next steps

This project changed how I think about AI in my discipline. I expected the hardest part to be the code; it was the data. Collecting 58 hand-woven pieces took four visits and many conversations, and my first result was misleading because of how I split the photos. The model is useful as a prompt for closer inspection, not as a judge. Next, I would widen the dataset to other emirates and weaving groups, test under indoor lighting, and co-design the interface with weavers so that the tool supports their own authentication practices.

✔️ TUTOR NOTE — the reflection

It is honest, specific and forward-looking: what surprised me, what went wrong, what I learned, what I would do next.

It links back to the research questions and the student's creative practice.

References

Formatted in the UCA Harvard style (UCA Library, n.d.). Accessed dates use DD/MM/YYYY.

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Buolamwini, J. and Gebru, T. (2018) 'Gender shades: intersectional accuracy disparities in commercial gender classification' In: *Proceedings of Machine Learning Research* 81 pp.77-91. At: https://proceedings.mlr.press/v81/buolamwini18a.html (Accessed 23/09/2026).

Cimpoi, M., Maji, S., Kokkinos, I., Mohamed, S. and Vedaldi, A. (2014) 'Describing textures in the wild' In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. Columbus, OH: IEEE. pp.3606-3613. At: https://openaccess.thecvf.com/content_cvpr_2014/html/Cimpoi_Describing_Textures_in_2014_CVPR_paper.html (Accessed 23/09/2026).

Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K. and Fei-Fei, L. (2009) 'ImageNet: a large-scale hierarchical image database' In: *2009 IEEE Conference on Computer Vision and Pattern Recognition*. Miami, FL: IEEE. pp.248-255.

Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J.W., Wallach, H., Daumé III, H. and Crawford, K. (2021) 'Datasheets for datasets' In: *Communications of the ACM* 64 (12) pp.86-92. At: https://doi.org/10.1145/3458723 (Accessed 23/09/2026).

He, K., Zhang, X., Ren, S. and Sun, J. (2016) 'Deep residual learning for image recognition' In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. Las Vegas, NV: IEEE. pp.770-778. At: https://doi.org/10.1109/CVPR.2016.90 (Accessed 23/09/2026).

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UCA Library (n.d.) *UCA Harvard referencing.* At: https://mylibrary.uca.ac.uk/referencing (Accessed 23/09/2026).

UNESCO (2025) *Al Sadu, traditional weaving skills in the United Arab Emirates.* At: https://ich.unesco.org/en/RL/al-sadu-traditional-weaving-skills-in-the-united-arab-emirates-02223 (Accessed 23/09/2026).

Yosinski, J., Clune, J., Bengio, Y. and Lipson, H. (2014) 'How transferable are features in deep neural networks?' In: *Advances in Neural Information Processing Systems 27 (NIPS 2014)*. pp.3320-3328.

✔️ TUTOR NOTE — the reference list

Alphabetical by first author's surname; every in-text citation appears here and vice versa.

Titles of books and journals are in italics; article titles are in single quotation marks; online sources include 'At:' and an accessed date.

⚠️ WATCH OUT — referencing errors

Citing 'Google Scholar' or 'Wikipedia' as the source instead of the actual paper · missing years or page ranges · mixing referencing styles · URLs with no accessed date.

Appendices

Appendix A · Declaration of AI use

Tool	What I used it for	How I checked it
Claude (Anthropic)	Explaining a fastai error message; suggesting search synonyms	Tested the fix myself; ran the searches myself; all writing is my own
Grammar checker in Word	Spelling and grammar	Accepted or rejected each suggestion

AI conversations are referenced in the UCA format, for example: Anthropic (2026) *AI conversation with Student, A.* 04/10/2026. At: [shareable URL].

Appendix B · Search log

Full list of search strings, dates, databases, results screened and reasons for inclusion or exclusion (not reproduced in this exemplar).

Appendix C · Consent form

Blank consent form and a record of signed forms (names withheld). Covers photography, use for model training, sharing under CC BY-NC 4.0, credit preference and the right to withdraw before 30/11/2026.

Appendix D · Model card and confusion matrix

Model card following Mitchell *et al.* (2019): intended use, not-intended uses, training data, evaluation data, metrics, ethical considerations and caveats (not reproduced in this exemplar).